

STATGRAPH: Effective In-Vehicle Intrusion Detection via Multi-View Statistical Graph Learning

Kai Wang[✉], *Member, IEEE*, Qiguang Jiang[✉], Bailing Wang[✉], *Member, IEEE*, Yulei Wu[✉], *Senior Member, IEEE*, and Hongke Zhang[✉], *Fellow, IEEE*

Abstract—In-vehicle networks (IVNs) face growing threats from advanced cyber-attacks, particularly stealthy masquerade attacks that mimic legitimate message patterns. This paper proposes STATGRAPH, a fine-grained intrusion detection framework based on multi-view statistical graph learning over the Controller Area Network (CAN) messages within IVNs. STATGRAPH constructs two graphs per detection window: a Timing Correlation Graph (TCG) capturing temporal ID dependencies, and a Coupling Relationship Graph (CRG) modeling short-term contextual relations. TCG and CRG are further used to generate graph structure encoding payload variations and embedded signal co-occurrence. A lightweight multi-layered Graph Convolutional Network (GCN) is then applied to classify each message, leveraging the expressive representations from TCG and CRG. To ensure effectiveness against diverse attacks, we evaluate STATGRAPH on two real-world CAN datasets featuring five underexplored masquerade attacks. Experimental results show that STATGRAPH significantly improves detection granularity and outperforms state-of-the-art methods, with F1-score gains of 7% and 22%, while maintaining the highest accuracy.

Index Terms—Intrusion detection, masquerade attacks, Internet of Vehicles, in-vehicle network, controller area network, multi-view statistical graph.

I. INTRODUCTION

AS A vital component of vehicle-to-everything (V2X) systems, intelligent connected vehicles (ICVs), which are

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Kai Wang and Qiguang Jiang are with the School of Computer Science and Technology, Harbin Institute of Technology, Weihai 264209, China, and also with the Shandong Key Laboratory of Industrial Network Security, Weihai 264209, China (e-mail: dr.wangkai@hit.edu.cn; jiangqiguang_971@163.com).

Bailing Wang is with the Harbin Institute of Technology (Weihai) Qingdao Research Institute, Qingdao 266109, China, and also with the Shandong Key Laboratory of Industrial Network Security, Weihai 264209, China (e-mail: wbl@hit.edu.cn).

Yulei Wu is with the Faculty of Engineering and the Bristol Digital Futures Institute, University of Bristol, BS8 1QU Bristol, U.K. (e-mail: y.l.wu@bristol.ac.uk).

Hongke Zhang is with the School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing 100044, China (e-mail: hkzhang@bjtu.edu.cn).

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equipped with in-vehicle communication networks and various embedded computing devices, have rapidly proliferated and become a prevailing trend [1]. With the increasing number of intelligent technologies integrated into vehicles, ICVs now offer a variety of advanced services and enhanced user experiences, such as autonomous driving, collision avoidance, and automated parking assistance. However, these technologies have also introduced new external attack surfaces to in-vehicle networks (IVNs), making them more vulnerable to a growing range of evolving and sophisticated cyber threats. These threats jeopardize not only the security of IVNs but also the safety of passengers [2].

Among in-vehicle communication protocols, the Controller Area Network (CAN) is widely regarded as the de facto standard for communication among Electronic Control Units (ECUs) in embedded IVN systems [3]. Malicious CAN messages can lead to severe consequences, particularly when they target critical systems such as the brakes, powertrain, or safety controls of a vehicle [4]. In light of this, recent studies have proposed various methods to enhance IVN security by detecting intrusions on the CAN bus [5], [6], [7]. While encryption and authentication techniques can enhance the security of IVNs [8], they often necessitate modifications to the existing CAN protocol specifications and communication procedures. Such changes are generally considered impractical and cost-prohibitive, given the widespread adoption of the CAN protocol in the automotive manufacturing industry. As a more feasible alternative, the intrusion detection system (IDS) offers a non-invasive security solution by leveraging bypass monitoring mechanisms that do not require any changes to the CAN protocol or ECU hardware. This characteristic makes IDS particularly suitable for real-world deployment, offering both technical viability and economic efficiency [9].

As artificial intelligence advances in network traffic identification, IVN intrusion detection has shifted from monitoring the physical characteristics of the ECU to analyzing the characteristics of the CAN data flow, reflecting interactions between the ECUs. Mainstream approaches predominantly use deep neural networks or graph-based methods, such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and graph-based intrusion detection systems [10], [11], [12]. However, existing supervised neural network models label all CAN messages within a data flow as anomalous, regardless of the number of actual abnormal messages [13]. This leads to coarse-grained detection, where only the overall anomaly of a

detection window is identified, without fine-grained recognition of individual messages [14], [15]. Consequently, the traceability of detection is lost, making it difficult to locate the exact source of anomalies [16], [17], [18]. Additionally, current graph-based methods primarily focus on simpler attack types [10], [19], leaving the detection of sophisticated masquerade attacks largely unexplored [9], [20], [21], [22]. Masquerade attacks are especially stealthy, as they replace legitimate messages while maintaining transmission timing and interaction patterns, but inject malicious payloads. However, most mainstream single-view graph-based intrusion detection methods primarily model the relationships among CAN message IDs and their transmission frequencies. While these approaches are effective for detecting simple anomalies, they often fail to identify masquerade attacks that modify only the payload, due to their limited ability to capture the coupling structure of data flow [11].

To address the aforementioned issues, we propose STATGRAPH, an effective in-vehicle intrusion detection approach that achieves fine-grained classification through multi-view statistical graph learning. Unlike previous single-view models, STATGRAPH adopts a dual-graph learning paradigm that simultaneously captures global statistical regularities and local contextual dependencies. It constructs node feature vectors and adjacency matrices for the Graph Convolutional Network (GCN) by generating two statistical views: the Timing Correlation Graph (TCG) and the Coupling Relationship Graph (CRG). The TCG models long-range statistical distributions among CAN messages by capturing temporal correlations between different CAN IDs within detection windows. In contrast, the CRG encodes both temporal adjacency and contextual similarity, reflecting the continuity of identical CAN IDs and variations in payload values or signal-level patterns over short time spans. These two graph views are integrated through graph convolutional operations, which enable multi-view feature propagation and abstraction. This design results in enriched representations of in-vehicle communication dynamics and enhances the model's ability to detect complex and subtle anomalies. Additionally, the GCN's compatibility with small-scale graphs in streaming contexts, along with its computational efficiency, makes it well suited for IVN intrusion detection scenarios where rapid response is essential.

The main contributions of this paper can be summarized as follows:

- We propose STATGRAPH, a multi-view statistical graph learning framework for effective intrusion detection in IVNs. By considering both the interactive relationships among CAN messages and their payload-level contextual similarities, STATGRAPH accurately captures variations in periodicity, payload content, and signal combinations. This enables the detection of a wide range of attacks, including sophisticated and complex masquerade attacks.
- Current deep learning-based approaches build the model input with multiple CAN messages, resulting in coarse recognition and the inability to locate the malicious message. To accurately quantify the message-level recognition accuracy, we introduce a new criterion, Identification Granularity (IG), which is used to assess how precisely



Fig. 1. CAN frame.

an intrusion detection model can classify each individual CAN message within a detection window, in contrast to traditional metrics that evaluate performance at the window or segment level.

- To evaluate the classification ability of STATGRAPH on captured CAN messages of realistic in-vehicle environment, we conduct experiments on an in-vehicle computing platform (NVIDIA Jetson Nano T206 with ARM architecture) and on a personal computer (LENOVO 90VA000JCP), respectively. Besides, we select the Car Hacking dataset and the ROAD dataset to cover as many types of attacks as possible, which are both directly generated from real vehicles on sale. Notably, to the best of our knowledge, we are the first to investigate 5 new types of masquerade attacks introduced by the ROAD dataset, which can cause incorrect working status of physical components within real vehicles (e.g., wrong speed indicator position by injecting fake wheel speed values) and bring in severe damage on both vehicles and passengers.

The remainder of this paper is organized as follows: Section II introduces the related background knowledge. Section III describes our proposed STATGRAPH. We show the experimental setup and results in Section IV followed by the discussion in Section V. Section VI summarizes existing representative IVN intrusion detection methods. We conclude the paper in Section VII.

II. BACKGROUND

In this section, we provide a brief introduction to the IVN CAN protocol and adversary types related to our STATGRAPH.

A. IVN CAN Details

CAN is the current de facto standard for IVN, which is widely used in the automotive industry to connect different ECUs in vehicles and support messaging between ECUs. There are four types of CAN frames: 1) Data frame; 2) Remote frame; 3) Error frame; 4) Overload frame. Fig. 1 illustrates the format of a CAN data frame, which is the default mode used for CAN message transmission. CAN data frames support payloads of up to 8 bytes and have an 11-bit arbitration ID (CAN ID) that can be extended to 29 bits.

The *arbitration ID* (CAN ID) plays a dual role in CAN communication: it governs message priority during bus arbitration and simultaneously serves as a semantic identifier. During arbitration, the message with the numerically lowest CAN ID is granted bus access, while others must defer until the bus is idle. Beyond arbitration, the CAN ID conveys content-specific semantics, as defined by the Original Equipment Manufacturer (OEM)-provided communication matrix. For instance, in several publicly available datasets, ID 0x45D is transmitted by the

powertrain control module every 0.01s to report gas and brake pedal positions, while ID 0x102 from the same module contains drivetrain-related information. These IDs exemplify how CAN messages are tied to specific modules and functions, with fixed payload formats and transmission intervals.

The *Data Field* (payload) contains up to 8 bytes and typically encodes multiple physical signals. Each signal is precisely defined by the communication matrix, which specifies its start bit, length in bits, scaling factor, and valid range. This bit-level packing enables compact and efficient transmission, with multiple quantities multiplexed into a single message. However, this design also introduces attack surfaces: adversaries may tamper with individual bits to subtly alter signal values while maintaining the overall message frequency and structure. Such fine-grained manipulations are often undetectable by coarse-grained detection models, highlighting the necessity of payload-level analysis. Additionally, the coexistence of various transmission patterns (e.g., periodic, event-triggered, or conditional) introduces complex temporal dependencies among CAN messages, as discussed in [23]. This structural complexity motivates the use of graph-based methods that can effectively model inter-message relationships across time and semantics.

B. Threat Model

ECUs employ bit-by-bit arbitration and explicit broadcasting to transmit CAN protocol messages, enabling fast and efficient data exchange. An adversary may access the CAN bus through the OBD-II port or exploit the telematics service of the OEM via remote network access. Based on the attacker's objectives, such intrusions are typically classified into three categories.

1) *Fabrication Attacks*. An adversary compromises an ECU and injects malicious messages with forged IDs and data fields into the CAN bus. This is a common and straightforward attack, easily executed. Despite this, all legitimate ECUs remain active, transmitting their original data. Fabrication attacks increase CAN bus traffic, leading to dynamic and distributed changes in communication behavior. Examples include Denial of Service (DoS) attacks, fuzzy attacks, and targeted ID attacks.

2) *Suspension Attacks*. In this type of attack, legitimate ECUs can be disabled by an adversary. Specifically, the targeted ECU is weakly compromised and loses the ability to transmit messages, impairing other ECUs that rely on its data to function properly. As a result, suspension attacks can affect not only the compromised ECU itself but also other receiving ECUs.

3) *Masquerade Attacks*. These attacks combine fabrication and suspension techniques, exhibiting highly sophisticated and stealthy behavior patterns. The adversary first disables the ECU intended for replacement, removing it from the CAN network. A fully compromised ECU then sends spoofed messages with the same ID at a realistic frequency, effectively masquerading as the original. In this advanced strategy, only the payload is subtly altered, while the behavioral characteristics of the target ECU are preserved. As a result, message conflicts are avoided, and communication dynamics remain unchanged, causing many intrusion detection methods to fail.

III. METHODOLOGY

In this section, we give the system overview of the STATGRAPH, and then present the details of its key components.

A. System Overview

STATGRAPH consists of the following three components, as illustrated in Fig. 2. In contrast to prior works such as [11], which rely on a single statistical view of CAN message IDs, STATGRAPH introduces a multi-view graph construction framework that jointly models global statistical trends and local contextual dependencies. This dual-graph architecture is designed to capture complementary aspects of in-vehicle communication behavior, enabling a richer and more nuanced representation of message dynamics. The TCG captures long-range behavioral patterns by aggregating messages with the same CAN ID, enabling the detection of anomalies in message periodicity and distributional shifts. Complementing this, the CRG preserves message-level granularity and constructs edges based on temporal adjacency and ID similarity, allowing the model to detect localized irregularities in payload content and inter-signal consistency. Their integration through graph convolution enables feature propagation across the two graph structures, rather than relying on simple feature concatenation. This approach reinforces the model's capacity to generalize from complex communication patterns.

This architectural design provides a structured foundation that encodes three critical dimensions of communication behavior: periodicity, payload semantics, and signal relationships. These aspects are not modeled independently but are jointly embedded in the node features and adjacency structures, equipping the system with strong inductive biases for fine-grained and interpretable anomaly detection.

- *Global Feature Fusion Module* (Section III-B) constructs the TCG and derives statistical behavioral features such as frequency and temporal regularity. These features, together with the raw payload bytes, are fused into node-level feature vectors that encode long-term communication patterns in IVNs. In the TCG, each unique CAN ID within the detection window is represented as a node. All messages associated with the same ID are aggregated to extract global statistical attributes that collectively reflect the periodicity and distributional characteristics of message transmissions. The resulting rich and structured node representations benefit downstream relational modeling in the CRG to further capture multiple dimensions of communication behavior, including timing consistency, payload semantics, and inter-signal relationships.

- *Association Enhancement Module* (Section III-C) constructs the CRG to generate adjacency matrices that encode fine-grained communication dependencies among CAN messages. Unlike conventional adjacency constructions that rely solely on message co-occurrence or ID similarity, the CRG models two complementary relational patterns: temporal adjacency between consecutive messages and semantic consistency among messages sharing the same CAN ID. Within each detection window, every individual CAN message is treated as a separate node. Edges are established either between temporally adjacent

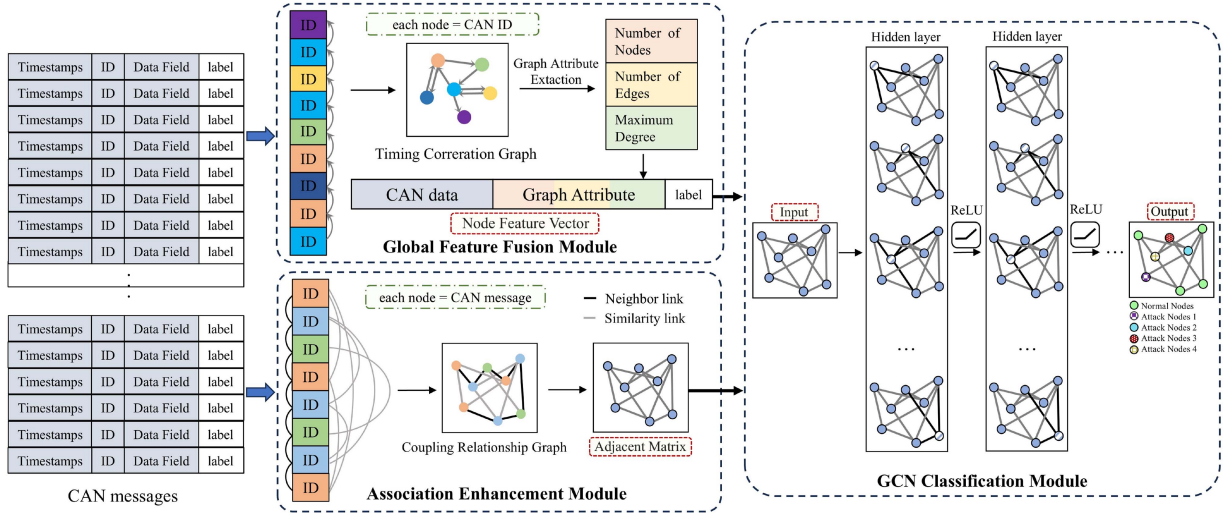


Fig. 2. Architecture of STATGRAPH. Global Feature Fusion Module (III-B) generates *node feature vectors* by constructing TCGs that react to the long-term distribution of IVN CAN messages. Association enhancement module (III-C) produces *adjacency matrices* by constructing CRGs whose edges reflect both contextual adjacency and intra-message dynamics between adjacent CAN messages or those with the same ID. GCN classification module (III-D) accepts the generation of streaming graph-structured data from the global feature fusion module and association enhancement module to perform training and detection tasks.



Fig. 3. NVIDIA Jetson Nano T206.

messages or between messages that share the same ID. This design enables the CRG to encode both temporal continuity and ID semantic regularities at the message level, thereby capturing localized structural patterns that reflect message periodicity, payload semantics, and inter-signal dependencies. These three dimensions are jointly embedded in the graph topology and serve as critical relational cues for detecting fine-grained anomalies that may elude traditional statistical aggregation methods.

- *GCN Classification Module* (Section III-D) consumes the graph-structured representations constructed by the Global Feature Fusion Module and Association Enhancement Module to perform message-level classification. The module takes as input the node features derived from the TCG, which reflect global statistical patterns such as frequency and temporal regularity, together with the adjacency matrices built from the CRG, which encode localized contextual and semantic relationships among CAN messages. This dual-graph input enables the GCN to jointly model long-term communication behavior and short-term structural dependencies, equipping it with strong inductive bias for detecting both overt and stealthy anomalies. By leveraging this architecture, the GCN module facilitates high-resolution threat detection by leveraging both statistical abstraction and structural semantics, thereby improving classification accuracy while preserving temporal and message-level interpretability.

We define two different types of generated multi-view in the following definition.

Definition 1: We consider two types of graphs: the TCG $G(V, E)$, which captures long-term statistical attributes, and the CRG $\hat{G}(\hat{V}, \hat{E})$, which models the topological structure of short-term message interactions. Here, $v_i, v_j \in V$ and $\hat{v}_i, \hat{v}_j \in \hat{V}$ denote nodes, while $e_{ij} \in E \subseteq V \times V$ and $\hat{e}_{ij} \in \hat{E} \subseteq \hat{V} \times \hat{V}$ represent edges. The values w_{ij} and $\hat{w}_{ij}$ are the respective weights of edges e_{ij} and $\hat{e}_{ij}$.

B. Global Feature Fusion Module

To capture long-term behavioral patterns in in-vehicle communication, this module constructs the TCG, which abstracts statistical dependencies among CAN IDs by extracting interaction features from continuous data streams. Unlike traditional approaches based mainly on frequency or co-occurrence statistics, the TCG employs directed edges to encode conditional dependencies derived from ID orderings within each detection window.

• Generation of TCG based on CAN stream

The CAN flow can be regarded as multivariate time series.

Definition 2: (Multivariate Time Series) CAN data is a multivariate time series $\Psi \in \mathbb{R}^{T \times V}$:

$$\Psi = \{\psi_1, \psi_2, \dots, \psi_t, \dots, \psi_T\} \quad (1)$$

where $\psi_t = \{\text{timestamp}, ID, DLC, d_1, \dots, d_8, \text{type}\} \in \mathbb{R}^V$. $V = 12$ is the dimension of a recorded data CAN message.

We choose $ID = d_0$, data field $\{d_1, d_2, \dots, d_8\} \in \mathbb{R}^8$ and label value d_l to form $\phi_t = \{d_0, d_1, d_2, \dots, d_8, d_l\} \in \mathbb{R}^{10}$ as input representation for STATGRAPH.

Since recurrent ID patterns reflect the periodicity of CAN messages, we divide IVN CAN traffic into multiple detection windows and characterize message distributions within each

window separately, enabling more efficient data preprocessing for STATGRAPH.

We divide the CAN detection windows as follows:

Definition 3: A series of CAN messages monitored from the IVN arranged according to the normal communication sequence can be regarded as located within a single detection window. In proposed algorithm, we consider N messages to be a detection window [11]. N is the detection window size. Let's assume that there are K detection windows in total. w_k denotes the matrix consisting of CAN messages within k -th detection window after preprocessing.

$$w_k = \{\phi_{k*N+1}, \phi_{k*N+2}, \dots, \phi_{k*N+N}\} \quad (2)$$

Therein, $k \in \{0, 1, 2, \dots, K-1\}$.

After dividing the CAN messages into K detection windows according to Definition 3, our scheme introduces Algorithm 1 to track valid transitions between the IDs of consecutive CAN messages and construct a TCG list based on multivariate time-series data. Specifically, for each set of N messages within a detection window, the algorithm treats each distinct CAN ID as a node and builds edges to represent ID-to-ID relationships. For clarity, $Set[ID_i] = \{\phi_n \mid \phi_n[d_0] = ID_i, \phi_n \in w_k\}$ denotes the set of messages with the same ID_i , where ID_i is the key for $Set[ID_i]$. The algorithm iterates over all detection windows (Line 2). For the k -th window, it collects each unique ID and creates a new key-value pair when the ID first appears (Lines 4 and 8). Then, an edge is added between the IDs of two consecutive messages. If the edge already exists, its weight is incremented (Line 11).

In the resulting TCG, each node corresponds to a unique CAN ID, and directed edges represent long-term statistical dependencies based on co-occurrence patterns observed within detection windows. Specifically, a directed edge from node ID_B to node ID_A is added whenever ID_B appears after ID_A within the same detection window, regardless of how frequently this occurs. In other words, the presence of such an edge does not imply a minimum frequency threshold, but rather reflects the existence of at least one instance of sequential co-occurrence. Each time ID_B follows ID_A , the weight of the edge from ID_B to ID_A is incremented by one, thus encoding the strength of this historical statistical association. This edge orientation does not represent a strict physical or chronological order, but instead models a backward statistical dependency: the current behavior of a CAN ID is understood in relation to the aggregate patterns of its preceding IDs. Consequently, the edge direction captures accumulated historical influence rather than deterministic temporal causality.

The rationale for this edge orientation is rooted in the assumption that the behavior of a CAN ID is influenced by the statistical characteristics of its predecessors. To capture this influence in a graph neural network, where information propagates from neighboring nodes to a target node during aggregation, the edge must be directed from ID_B to ID_A . This configuration allows node ID_B to receive contextual information from node ID_A , thereby encoding historical dependencies into the node representations. Such a design is particularly effective in identifying masquerade attacks, where adversarial messages mimic the surface statistics of normal traffic but disrupt underlying

Algorithm 1: TCG Building Algorithm.

Require:

Detection Window List of CAN Messages:

$CANWindowList = \{w_1, w_2, \dots, w_K\}$, therein $w_k = \{\phi_{k*N+1}, \phi_{k*N+2}, \dots, \phi_{k*N+N}\}$.

Ensure:

$CorrelationGraphList[G_1, G_2, \dots, G_K] \triangleright$ Timing

Correlation Graph array of CAN bus data;

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1:  $CorrelationGraphList \leftarrow []$ ,  $IDDictionary \leftarrow \{\}$ 
2: for  $w_k$  in  $CANWindowList$  do
3:   Initialize  $Graph$ ;
4:    $LastID \leftarrow \phi_{k*N+1}[d_0]$ ;
5:    $IDDictionary[LastID] \leftarrow len(IDDictionary)$ 
    $\triangleright$  Create a new key/value pair in the dictionary;
6:   for  $i$  in  $\{2, \dots, N\}$  do
7:      $NowID \leftarrow \phi_{k*N+i}[d_0]$ ;
8:     if not  $NowID$  in  $IDDictionary.keys()$  then
9:        $IDDictionary[NowID] \leftarrow len(IDDictionary)$ 
        $\triangleright$  Create a new key/value pair in the dictionary;
10:    end if
11:    Connect an edge from  $v_{IDDictionary[NowID]}$  to  $v_{IDDictionary[LastID]}$ 
     $\triangleright$  Create link between two graph nodes;
12:     $LastID \leftarrow NowID$ ;
13:  end for
14:  Append  $Graph$  to the  $CorrelationGraphList$ ;
15:   $IDDictionary.clear$   $\triangleright$  clear  $IDDictionary$ 
16: end for
```

transmission patterns. By modeling statistical dependencies in this manner, the TCG enhances the model's capacity to recognize subtle anomalies in message sequences.

$$w_{i,j} \leftarrow \begin{cases} +1, & \phi_n \in Set[ID_j], \phi_{n+1} \in Set[ID_i] \\ +0, & \text{else} \end{cases}$$

The algorithm keeps the graph G_k built by the current detection window (Line 14) and eventually returns the graphs $G_1, \dots, G_K$ for all detection windows.

- *Building feature vector of each node*

After construction of TCG, graph feature vectors can be generated by statistical attributes of TCG, such as: the number of edges, the number of nodes, the max weight, the average weight, the max degree, etc. Our proposed method characterizes feature vector for each CAN message, which could be defined as $X_{k,n}$, denoting the node feature of n -th message in k -th detection window of raw CAN messages, $n \in \{1, 2, \dots, N\}$. Hence, denote node feature matrixes of k -th detection window as:

$$X_k, k \in \{0, 1, \dots, K-1\} \quad (3)$$

According to research [11], we choose to extract the node number, edge number, and maximum degree from a single constructed TCG, and consider them as global features to be appended to the node feature vectors.

Hence, node feature is designed accordingly as $X_{k,n} = \{d_0, d_1, d_2, \dots, d_8, x_9, \dots, x_{11}\} \in \mathbb{R}^{12}$, where d_0 is CAN ID converted to decimal, $d_1, d_2, \dots, d_8$ mean the data value in data field and x_9, x_{10}, x_{11} represent the node number, edge number and maximum degree of the TCG to which the node belongs, respectively. To test and verify the detection granularity for each CAN message, the node feature $X_{k,n}$ retains the label in the model training and testing process indicating whether it is normal or injected, though it is not a necessary part of the input node feature vector when implemented in the real IDS environment.

The TCG models the periodic transmission behavior of CAN messages based on the timestamps and IDs of consecutive entries. By analyzing ID transitions and their recurrence within detection windows, the TCG captures temporal regularities and detects deviations indicative of anomalous behavior. As legitimate messages typically exhibit stable timing patterns, disruptions caused by injected messages can be identified through statistical variations such as frequency shifts and entropy fluctuations. For example, a message with ID 0x45D, normally transmitted every 10 milliseconds, may display altered periodicity under attack conditions.

In addition, the TCG provides statistical features to construct node feature vectors, which incorporate payload bytes d_1 to d_8 along with temporal statistics, forming a representation that captures both content and timing characteristics. This design integrates message periodicity, payload content, and signal correlations into a unified node feature representation for the first time, yielding a more expressive global view and offering a stable, generalized structural prior for downstream relationship modeling.

C. Association Enhancement Module

This module leverages CRGs to enhance the association between CAN messages that exhibit similarity and short-term dependence. While TCGs capture periodicity and regularity, they cannot detect masquerade attacks, which alter payloads without affecting statistical patterns. Legitimate message payloads fluctuate but exhibit regularities within the same ECU [24], and CAN messages from different ECUs may share common patterns, which masquerade attacks disrupt.

CRGs capture these coupling relationships by reinforcing links between messages based on short-term correlations. Unlike approaches that rely solely on contextual connections due to the simplicity of the CAN format, CRGs integrate both contextual correlation and content similarity, providing a more comprehensive representation of CAN message associations. As illustrated in Fig. 2, the gray links denote similarity relationships among groups of CAN IDs, whereas the black links capture the contextual correlation between successive CAN messages.

Despite substantial efforts to model graph structural information, existing methods remain limited in capturing coupled relationships [13]. By incorporating both contextual adjacency (the contextual relevance among two adjacent messages) and intra-message dynamics (the similarity of intra message with the same ID), the CRG preserves localized communication patterns

Algorithm 2. CRG Building Algorithm.

Require:

Detection Window List of CAN Messages:

$CANWindowList = \{w_1, w_2, \dots, w_K\}$, therein $w_k = \{\phi_{k*N+1}, \phi_{k*N+2}, \dots, \phi_{k*N+N}\}$.

Ensure:

$RelationshipGraphList[\hat{G}_1, \hat{G}_2, \dots, \hat{G}_K] \triangleright$ Coupling

Relationship Graph array of CAN bus data;

1: $RelationshipGraphList \leftarrow [], IDSet \leftarrow \{\}$

2: **for** w_k in $CANWindowList$ **do**

3: Initialize $Graph$;

4: $LastID \leftarrow \phi_{k*N+1}[d_0]$;

5: $IDSet[LastID] \leftarrow len(IDSet) \triangleright$ Create a new key/value pair in the dictionary;

6: **for** i in $\{2, \dots, N\}$ **do**

7: $NowID \leftarrow \phi_{k*N+i}[d_0]$;

8: $(\hat{v}_{i-1}, \hat{v}_i) \leftarrow 1 \triangleright$ Link two neighbor nodes (Contextual correlation property);

9: **if** not $NowID$ in $IDSet.keys()$ **then**

10: $IDSet[NowID] \leftarrow [i] \triangleright$ Create a new key/value pair in the dictionary;

11: **else**

12: Append i to the $IDSet[NowID]$;

13: Link the edges between nodes in $IDSet[NowID] \triangleright$ Connect all nodes with the same ID value (Similarity);

14: **end if**

15: **end for**

16: Append $Graph$ to the $CorrelationGraphList$;

17: $IDSet.clear \triangleright$ clear $IDSet$

18: **end for**

and content semantic consistency at the message level, which facilitates the detection of subtle variations in payload content and embedded signals in each CAN message. In STATGRAPH, the CRG innovatively defines data flow coupling by capturing two essential forms of contextual correlation: temporal adjacency between consecutive CAN messages and semantic consistency among messages sharing the same ID. Unlike conventional methods that construct graphs based only on message co-occurrence or ID similarity, the CRG preserves message-level granularity and incorporates both temporal ordering and ID relationships into its adjacency structure. This design enables the model to retain fine-grained contextual dependencies, allowing it to detect localized anomalies such as unexpected changes in payload values or disrupted signal correlations embedded in d_1 to d_8 . The CRG thus offers a structural representation of communication patterns that are sensitive to violations in temporal continuity, payload integrity, and inter-signal coherence. To operationalize these relationships, we propose a novel construction strategy for the CRG's adjacency matrix, as detailed in Algorithm 2.

From Definition 3, we set the same detection windows as Section III-B. Each CAN message in the k -th detection window is treated as a node in the CRG, which contains N nodes, and edges

are constructed to represent short-term relationships between messages. A loop is iterated over each detection window (Line 2). In the k -th detection window, we define the nodes with the same ID as a set denoted $Set[ID_i] = \{\phi_n | \phi_n[d_0] = ID_i, \phi_n \in w_k\}$, which is constructed by counting the messages corresponding to each ID (Lines 5 and 9). Next, the algorithm constructs edges of CRG $\hat{G}_k$ by searching coupling relationships. For i, j in $\{1, 2, \dots, N\}$, we have

$$\hat{e}_{ij} = \begin{cases} 1, & j = i + 1 \text{ or } i = j + 1 \\ 1, & \phi_j, \phi_i \in Set[ID_m] \\ 0, & \text{else} \end{cases} \quad (4)$$

The first row in (4) specifies that an edge is created between nodes representing two consecutive CAN messages (Line 7). The second row in (4) indicates that nodes sharing the same CAN ID are interconnected (Line 12). The last row of (4) states that edges not satisfying the aforementioned relationships are excluded from the graph.

Finally, Algorithm 2 returns undirected graph instances of CRGs for multiple detection windows, whose adjacency matrices will be used for the classification task in the GCN by learning generalizable patterns. Therefore, the adjacency matrix of k -th detection window as input for next GCN classification module can be defined as:

$$A_k, k \in \{0, 1, \dots, K - 1\} \quad (5)$$

The CRG captures short-term communication dependencies by treating each individual CAN message as a node and constructing edges based on temporal adjacency and ID similarity. Edges are constructed to reflect the expected local communication flow and semantic regularities. This graph is built directly from message timestamps and CAN IDs, enabling the model to represent fine-grained variations in message timing. Subtle temporal deviations such as jitter or irregular intervals may signal stealthy intrusions and can thus be effectively identified. Although the CRG does not explicitly interpret the semantics of the payload, it retains payload similarity structures among messages, including correlations across the eight signal bytes d_1 through d_8 . This retention allows the model to capture changes in contextual consistency and signal co-evolution patterns. For instance, messages with the same ID but exhibiting sudden shifts in payload distributions or disrupted signal correlations may indicate semantic manipulation or low-rate injection attacks. By preserving message-level granularity and local structural continuity, the CRG transforms subtle inconsistencies into structural variations, which can be effectively captured by the downstream GCN classifier.

D. GCN Classification Module

We propose a multi-layer GCN model for classifying attacks on the in-vehicle CAN bus, trained using graph-structured data generated by the Global Feature Fusion and Association Enhancement modules, which capture both temporal and contextual dependencies of IVN message streams.

Based on this design, the GCN module functions as the core classifier in STATGRAPH, integrating node-level features from

TABLE I
STRUCTURE AND DIMENSIONS OF ALL LAYERS OF STATGRAPH

Role	Layer Name	Data Shape	Activation	Dropout Ratio
Input	Input	$(B*N) \times 12$	-	-
	GCN 1	$(B*N) \times h$	ReLU	-
Feature Extraction	Dropout 1	$(B*N) \times h$	-	0.5
	GCN 2	$(B*N) \times h$	ReLU	-
	Dropout 2	$(B*N) \times h$	-	0.5
	GCN 3	$(B*N) \times h$	ReLU	-
	...			
	Dropout $l - 1$	$(B*N) \times h$	-	0.5
	GCN l	$(B*N) \times h$	ReLU	-
	...			
	Dropout $L - 1$	$(B*N) \times h$	-	0.5
	GCN L	$(B*N) \times h$	ReLU	-
Multi-Classification	Dense	$(B*N) \times F$	Sigmoid	-

the TCG with edge-level structures defined by the CRG. The TCG models long-range communication patterns by aggregating messages with the same CAN ID, enabling the extraction of global statistical features such as frequency shifts and periodic anomalies. In contrast, the CRG preserves message-level granularity and constructs the adjacency matrix by jointly modeling two types of relational dependencies: temporal adjacency among consecutive messages and semantic similarity among messages sharing the same CAN ID. This allows the CRG to encode both the contextual flow and ID structural regularities. As a result, disruptions in expected message timing, payload consistency, or inter-signal correlations can be reflected as abnormal edge patterns. By jointly leveraging the statistical aggregation of TCG and the localized structural encoding of CRG, the GCN receives a rich representation of communication behavior, which supports precise and fine-grained classification of each message. This unified modeling approach ensures that both global distributional shifts and localized anomalies are effectively captured. The GCN model [25] applies first-order graph convolution to jointly model structural information and node attributes.

To formalize the learning process, the multi-layer GCN is defined as:

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)} \right)$$

where $H^{(l)}$ is the matrix of activations in the l -th layer, $W^{(l)}$ denotes the trainable weight matrix of l -th layer. $\sigma(\cdot)$ is the activation function. Denote number of hidden layer units as h , and $l \in \{0, 1, 2, \dots, L\}$. Set the adjacency matrix of input graph as A . $\tilde{A} = A + I_M$, where M is the size of A and I_M is the M -dimensional identity matrix, $\tilde{D}$ is degree matrix of $\tilde{A}$ and $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Denoting the node feature matrix of input graph as X , we can get initial $H^{(0)} = X$. After multiple experiments, from the perspective of lightweighting and performance, our GCN model consists of an input layer, L hidden layers and a dense layer, shown in Table I.

In order to improve the training efficiency of the model in coding, we set $batchsize=B$, which means that we take the node feature matrixes and the adjacency matrixes corresponding to B consecutive detection windows as once input. Hence, the input node feature matrix $X \in \mathbb{R}^{(B*N) \times 12}$ is the continuous X_k splice of B and the input adjacency matrix A with size $(B * N, B * N)$ is a diagonal concatenation of B continuous matrixes A_k .

TABLE II
HYPERPARAMETERS FOR TRAINING THE STATGRAPH MODEL

<i>Hyperparameters</i>	Value(Car Hacking)	Value(ROAD)
B (Batchsize)	40	5
N (Detection Window Size)	50	400
L (Total Number Of Layers)	4	1
h (Number of Hidden Layer Units)	32	32
F (Number of Categories)	5	6
Learning Rate	5×10^{-4}	5×10^{-4}
Weight Decay	5×10^{-4}	5×10^{-4}

With the structural and feature information embedded in the input, the GCN proceeds to node classification, producing abstract representations for each CAN message to enable fine-grained detection.

For k -th detection window, the last dense layer of GCN model returns output $Z_k = \{Z_{k,1}, \dots, Z_{k,N}\}$, where Z_k is the set of $Z_{k,n}$ and $Z_{k,n} \in \{0, 1, 2, \dots, F\}$ is employed to predict the attribute labels of the n -th message in the k -th detection window. In the training phase, the *cross-entropy* loss function compares real label of the n -th message in the k -th detection window to $Z_{k,n}$ through a sigmoid activation function.

Besides, L2 regularization is used to restrict the values of weights to prevent overfitting. Besides, our GCN model is trained using the Adam optimizer, with a learning rate of 10^{-3} . In addition, we deploy dropout layers with a rate of 0.005 after each GraphConv layer. Moreover, we set the rectified linear unit (ReLU), a widely used nonlinear function, as activation function in our model. It introduces nonlinearity and allows the neural network to learn complex relationships in the data. And hyperparameters for training the STATGRAPH are shown in Table II.

This module fully utilizes node features by fusing statistical regularities derived from recurrent ID patterns and association relations enhanced by adjacency matrices, thereby enabling high-performance, fine-grained detection and classification of multiple attacks. The use of backward edge direction in the TCG is motivated by the GCN message aggregation mechanism, where each node aggregates information from its incoming neighbors. By directing edges from the latter ID to the former, the model allows emerging nodes to incorporate historical contextual features during convolution, facilitating the learning of temporal correlations among CAN IDs, particularly those with fixed or periodic transmission characteristics. By jointly leveraging both the TCG and CRG, the STATGRAPH framework captures multiple dimensions of in-vehicle communication dynamics. The TCG constructs node features by aggregating all messages sharing the same CAN ID within a detection window. This process extracts statistical characteristics such as frequency, entropy, and inter-arrival time, enabling the encoding of long-term communication patterns. Directed edges in the TCG represent statistical dependencies among different IDs, allowing the GCN to aggregate contextual information from historically correlated message flows. The CRG, in contrast, defines the adjacency matrix based on two criteria: temporal adjacency between consecutive messages and ID similarity. This graph preserves message-level granularity and captures localized contextual and semantic dependencies, including fine-grained timing variations and signal correlations inferred from payload bytes.

Together, the global and local priors encoded by the TCG and CRG enable joint representation learning that effectively models multi-scale behavioral patterns in in-vehicle communication. The dual-graph design produces highly expressive inputs for GCN-based classification, allowing even shallow architectures to achieve strong performance. This efficiency stems from the rich feature encoding in the graph representations, reducing the need for complex model depth while enabling robust detection of stealthy intrusions, such as masquerade attacks that preserve surface-level message patterns.

IV. EXPERIMENT

This section presents the implementation of the STATGRAPH framework under two computing environments (Section IV-A) and introduces the evaluation metrics used to quantify detection granularity (Section IV-B). Baseline methods for comparison are described in Section IV-C. The dataset details are provided in Section IV-D, followed by a parameter sensitivity analysis in Section IV-E. We evaluate runtime and detection performance in Sections IV-F and IV-G, respectively. The fine-grained detection capability is explored in Section IV-H.

A. Environment

As a core technology for vehicle network security, the intrusion detection model must offer real-time detection capabilities and a low false positive rate, particularly in vehicle environments with limited computational resources.

With this in mind, we deploy and evaluate it on the NVIDIA Jetson Orin Nano platform (as shown in Fig. 3), a widely adopted hardware system in both the automotive industry and academia. This platform supports CAN bus communication, features an ARMv8 Processor rev 1, and integrates a 512-core NVIDIA Ampere architecture GPU with 8.0 GB of memory. With approximately 21 Tera Operations Per Second (TOPS) of computing power and low energy consumption, the device supports inference for medium-sized deep learning models, including tasks such as target detection and time-series data analysis, making it well suited for evaluating complex intrusion detection algorithms under resource-constrained onboard conditions. The software environment includes PyTorch 1.12.1 with Python 3.8.10.

For training, we utilize a more powerful device, the LENOVO 90VA000JCP with 49 TOPS of computational power, to simulate a cloud environment and accelerate the training process. It features a 64-bit Intel Core i7-13700 CPU at 3.6 GHz and a Geforce RTX 4080 32 GB GPU, with PyTorch 1.13 and Python 3.8 for the STATGRAPH model.

B. Evaluation Metrics

To evaluate the effectiveness of different methods, this paper adopts a set of standard classification metrics. Accuracy, Precision, and Recall can be calculated using the numbers of true negatives (TN), true positives (TP), false negatives (FN), and false positives (FP), defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

Besides, F1-score is more useful than accuracy in the case of an unbalanced class distribution. It is the harmonic mean of precision and recall, calculated as follows:

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

IG is a fine-grained evaluation metric that quantifies the message-level recognition accuracy of intrusion detection methods. This metric is particularly necessary because existing graph-based intrusion detection approaches for IVNs predominantly adopt coarse-grained classification strategies, wherein a single anomaly label is typically assigned to an entire message window or sequence. Such granularity is often insufficient for timely and precise threat localization, especially in scenarios involving stealthy or transient attacks. To address this limitation, STATGRAPH is, to the best of our knowledge, the first graph-based framework specifically designed to enable fine-grained, message-level intrusion detection. By providing classification results for each individual CAN message, the proposed framework achieves higher temporal resolution and improved diagnostic accuracy. The IG metric is introduced to formally quantify this capability, as it captures evaluation aspects that are overlooked by conventional, window-level performance metrics.

Based on the correct identification of the detection window (that is, the label of the k -th detection window, $label_k$ = the predicted value of the k -th detection window, $predict_k$), IG is calculated through TI (True Identification), which represents the number of correctly identified samples per detection window. Hence, the IG is defined as follows:

$$\text{IG} = \frac{1}{N} \sum_{k=1}^K \text{TI}_k \times C_k \quad (10)$$

where

$$C_k = \begin{cases} 1, & label_k = predict_k \\ 0, & label_k \neq predict_k \end{cases}$$

It measures how precisely a model can determine the status of each individual CAN message within a detection window, thus enabling a clear distinction between models that perform well only at a coarse-grained level and those capable of delivering reliable fine-grained identification.

C. Compared Methods

To ensure a fair comparison, several representative and advanced models are selected for binary and multi-classification tasks, as well as for coarse-grained and fine-grained levels:

- *Graph-based intrusion detection system (Graph-based IDS)* [11]: A coarse-grained, graph-based IVN intrusion detection method that uses the median and Chi-squared tests for binary classification tasks. It operates on the

TABLE III
OPEN CAN DATASETS

Dataset	Real/ Synthetic	Fabrication Attack	Masquerade Attack
CAN Intrusion Dataset(OTIDS) [30]	Real	Real;Dos;Fuzzing	-
Survival Analysis Dataset [31]	Real	Real;Dos;Fuzzing; targeted ID	-
Car Hacking Dataset [13], [32]	Real	Real;Dos;Fuzzing; targeted ID	-
SynCAN Dataset [16]	Simulated	Simulated; targeted ID	Simulated
Automotive CAN Bus Intrusion Dataset v2 [33]	Real	Simulated; targeted ID	Simulated
Can Log Infector	Real	-	Provide 7 types codes to simulate
ROAD Dataset [19]	Real	Real;Fuzzing; targeted ID	Simulated

number of nodes, edges, and maximum degree of graph-structured data generated from CAN messages over six detection windows.

- *G-IDCS TH_classifier* [12]: This graph-based IVN intrusion detection method performs binary classification based on thresholds for the number of nodes, edges, and maximum degree of graph-structured data within each detection window.
- *G-IDCS ML_classifier* [12]: Using the same graph generation process as G-IDCS TH_classifier, this method performs multi-class classification tasks using a Random Forest (RF) algorithm on the generated graph-structured data within each detection window.
- *EfficientNet* [26]: A neural network architecture with a novel compound scaling method that balances network width, depth, and resolution. It performs coarse-grained multi-classification intrusion detection tasks in resource-constrained IVN environments.
- *MobileNetV3* [27]: A lightweight neural network that utilizes efficient depthwise separable convolutions, suitable for IVN multi-classification intrusion detection tasks at a coarse-grained level (window-level classification).
- *CANet* [16]: The first deep learning-based method for IVN multi-classification intrusion detection, which checks individual CAN messages using a tailored network structure that fits the signal space of CAN data for each detection window.
- *CAN-RF* [28]: A fine-grained multi-classification method based on an integrated learning framework where each decision tree is constructed from a random subset of features of each CAN message.
- *MultiLayer Perceptron (CAN-MLP)* [28]: This model executes multi-class classification tasks on each CAN message using a feedforward neural network structure.
- *Long Short Term Memory (CAN-LSTM)* [29]: A popular RNN variant that predicts the next CAN ID more effectively based on previous observations, making it suitable for fine-grained multi-classification tasks.

D. Datasets Details

To the best of our knowledge, seven publicly available CAN datasets with labeled attacks exist (see Table III). We prioritize

TABLE IV
THE DISTRIBUTION OF THE CAR HACKING DATASET

Data Type	Normal Data Quantity	Attack Data Quantity	Attack Data Proportion	Total Data
Normal	988,987	0	0%	988,987
DoS Attack	3,078,250	587,521	16%	3,665,771
Fuzzy Attack	3,347,013	491,847	13%	3,838,860
GEAR Attack	3,845,890	597,252	13%	4,443,142
RPM Attack	3,966,805	654,897	14%	4,621,702

datasets involving real attacks on actual vehicles, as simulated data may lack real-world fidelity. While prior IDS research has largely focused on fabrication attacks, masquerade attacks remain underexplored. The Car Hacking dataset, widely used in IVN intrusion detection research, contains various fabrication attacks. In contrast, the ROAD dataset offers high fidelity and includes diverse masquerade attacks, with their physical effects validated on real vehicles. For example, injecting false wheel speed values can trigger critical malfunctions, posing safety risks to both the vehicle and passengers. Detailed descriptions and experimental setups for each dataset are presented in the following subsections.

1) *Car Hacking Dataset*: It was produced by the Hacking and Countermeasure Research Lab (HCRL) of Korea University, recording CAN traffic through the OBD-II port of a real vehicle [13]. It includes one normal dataset and four attack datasets: DoS, Fuzzy, spoofing GEAR, and spoofing RPM. The sample distribution of the Car Hacking dataset is presented in Table IV. For detection window integrity, we classify each window based on its content: a window containing both normal and attack data is categorized as an attack window, with labels for both malicious and normal frames. A window containing only normal data is classified as a normal detection window.

To ensure a balanced and non-overlapping dataset division, we first partitioned the normal samples in the *normal.csv* file by allocating 80% to the training set and the remaining 20% to the validation set. For attack samples contained in files such as *DoS.csv*, *Fuzzy.csv*, *GEAR.csv*, and *RPM.csv*, we assigned 70% to the training set, 20% to the validation set, and the remaining 10% to test the detection ability of STATGRAPH on unknown attacks, since they never appeared during model training. In addition to these attack samples, the test set also includes residual normal samples that were not utilized during training or validation, i.e., normal instances embedded within the attack files. This partitioning strategy maintains a representative distribution of normal and attack behaviors across all stages, which is particularly important given the large volume of normal messages in the dataset. It reflects real-world conditions, where normal traffic dominates even in the presence of attacks, and ensures clear separation between subsets. The same division is consistently applied across all models to support fair comparison and reproducibility.

2) *Road Dataset*: It includes a variety of CAN attacks collected from a passenger vehicle [19]. We choose the following five advanced masquerade attacks whose sample distribution is shown in Table V:

TABLE V
THE DISTRIBUTION OF REPRESENTATIVE MASQUERADE ATTACKS IN ROAD DATASET

Data Type	Normal Data Quantity	Attack Data Quantity	Attack Data Proportion	Total Data
Correlated Signal Masquerade Attack	167,974	5,453	3.14%	173,427
Max Speedometer Masquerade Attack	508,205	11,624	2.23%	519,829
Reverse Light Off Masquerade Attack	274,830	5,440	1.94%	280,270
Reverse Light On Masquerade Attack	415,239	7,930	1.87%	423,169
Max Engine Coolant Temp Masquerade Attack	57,097	43	0.075%	57,140

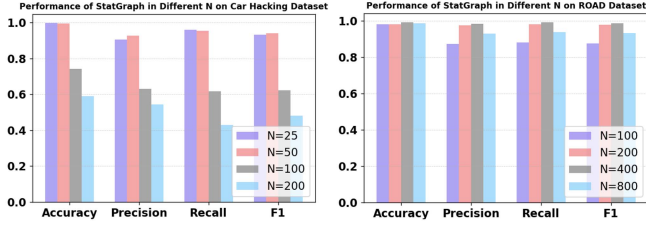
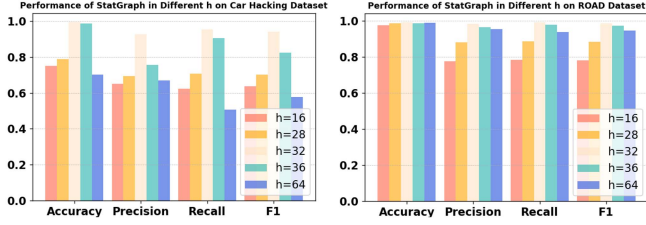
- *Correlated Signal Masquerade Attack*: The vehicle receives four false wheel speed values from injected malicious CAN messages, which disables the accelerator pedal and even makes a restart of the automotive electronic and electrical systems.
- *Max Engine Coolant Temp Masquerade Attack*: The vehicle receives an alarm of “engine coolant too high” that may mislead drivers to operate unnecessarily, due to the malicious modification of the engine coolant signal value with the maximum (0xFF) by attackers.
- *Max Speedometer Masquerade Attack*: The speedometer incorrectly displays the maximum value (0xFF) due to the injected malicious CAN messages, and may cause the driver to brake urgently.
- *Reverse Light Off Masquerade Attack*: The reverse lights of the vehicle are maliciously turned off when in reverse-gear, which may have danger to pedestrians due to lack of signal light alerts.
- *Reverse Light On Masquerade Attack*: The reverse lights of the vehicle are maliciously turned on when in drive-gear, which may mislead the vehicles following behind and result in improper operation.

The ROAD dataset provides both labeled signals translated using the CAN-D algorithm [34] and raw, unlabeled data. Since OEMs of passenger vehicles keep their proprietary CAN signal encodings confidential and vary them across models, the CAN-D algorithm is not always reliable for translating unknown CAN signals. The translation process only involves the payload, excluding the CAN ID and timestamps.

To address this, we compare the IDs and timestamps of the translated signals (in CSV format) with those of the raw data (in log files), considering messages with identical IDs and timestamps as corresponding messages. These are then formatted to match the structure of the Car Hacking Dataset.

E. Parameters Analysis

We evaluate the effects of different parameters that affect the performance of STATGRAPH to find the optimal detection window size N , number of hidden layer units h , and batchsize B . The experiments are conducted as follows:

Fig. 4. Performance of STATGRAPH in Different Detection Window Size N .Fig. 5. Performance of STATGRAPH in Different Hidden-layer Units h .

1) *Sensitivity to detection window size*: Fig. 4 illustrates the performance comparison for detection window sizes N ranging from 25 to 200 (Car Hacking) and 100 to 800 (ROAD). The results show that the impact of N is dataset-dependent. On the Car Hacking dataset, the detection window size significantly influences accuracy due to the simplicity and abrupt nature of the attacks, where small changes in N can alter the detection outcomes noticeably. The highest F1-Score is achieved with $N = 50$. In contrast, on the ROAD dataset, which contains more stealthy and complex attacks mimicking normal behavior, the detection performance is relatively stable across a wide range of window sizes. This indicates that once N is sufficiently large to capture the temporal and statistical patterns, further increases do not dramatically affect detection. The best F1-Score on ROAD is achieved with $N = 400$.

These findings imply that while the model leverages N to construct both TCG and CRG structures, it maintains superior performance when N is reasonably configured to align with the dataset's temporal characteristics.

This robustness can be partly attributed to the structural design of STATGRAPH. The TCG captures temporal rhythm changes through edge weight and node degree distributions, rather than relying on absolute message indices, while the CRG encodes short-term dependencies through local contextual relationships. Consequently, even under fluctuating message frequencies, structural anomalies remain detectable. Empirical results show that abrupt attacks in the Car Hacking dataset exhibit higher sensitivity to N , whereas stealthy attacks in the ROAD dataset benefit from larger windows that accumulate sufficient statistical deviations. These results demonstrate that the choice of N is closely related to the nature of attacks and highlight the importance of proper parameter configuration for practical deployment.

2) *Sensitivity to number of hidden layer units h* : Fig. 5 compares performance across different latent space dimensions, ranging from 16 to 64. The results reveal that excessively large dimensions lead to a significant decline in F1-Score, likely due

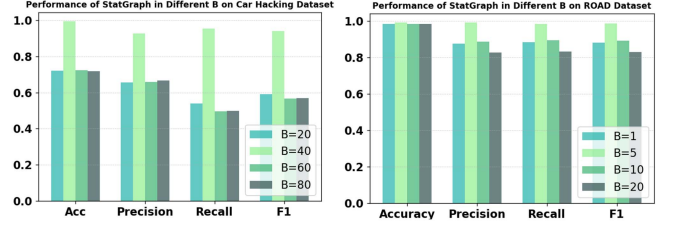
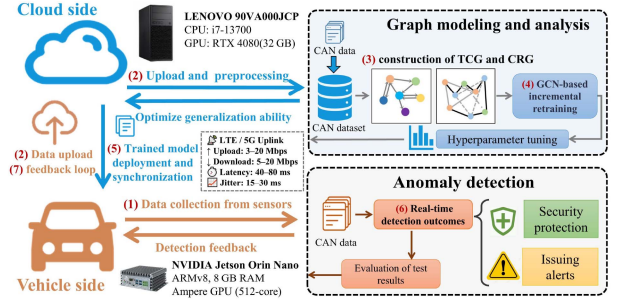
Fig. 6. Performance of STATGRAPH in Different Batchsize B .

Fig. 7. Experimental Flowchart of Vehicle-Cloud Detection Architecture.

to overfitting. Conversely, overly small dimensions result in substantial information loss during encoding, severely degrading GCN performance. A latent space dimension of 32 achieves the best F1-Score on both datasets, enabling STATGRAPH to balance training efficiency and accuracy.

3) *Sensitivity to batch size*: Fig. 6 compares performance across batch sizes from 20 to 80 on the Car Hacking dataset and from 1 to 20 on the ROAD dataset. The results indicate that both excessively large and small batch sizes degrade intrusion detection performance. The highest F1-Score is achieved with a batch size B of 50 for the Car Hacking dataset and 5 for the ROAD dataset, respectively.

F. Vehicle-Cloud Collaboration Framework

In real-world deployment, in-vehicle IDS must operate under strict resource constraints while ensuring real-time responsiveness and high detection accuracy. To meet these challenges, we adopt a vehicle-cloud collaborative framework that decouples training and inference, assigning them to the cloud and vehicle sides respectively for improved efficiency and robustness. Specifically, the cloud server performs offline training and optimization, while the vehicle-side platform (e.g., NVIDIA Jetson Nano T206) handles real-time inference. As shown in Fig. 7, the interaction between cloud and vehicle is bidirectional and continuous:

On the cloud side, a centralized platform handles model training and optimization by aggregating CAN message streams from multiple vehicles. It performs preprocessing and iteratively improves the model through incremental retraining with new attack patterns, as well as hyperparameter optimization. The trained model is then deployed to vehicle-side devices. In practice, dataset aggregation, model training, validation, and hyperparameter tuning are performed in the cloud, while the

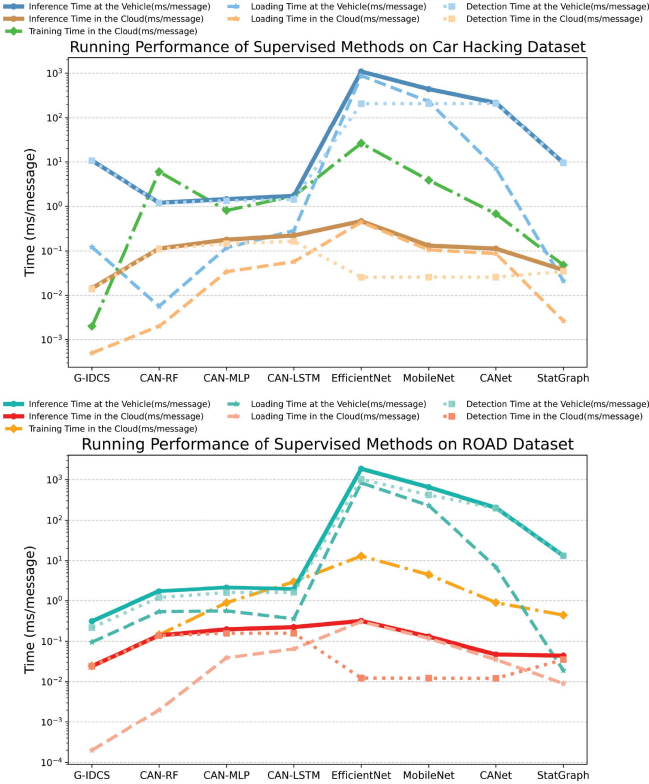


Fig. 8. Running Time Comparison Evaluation.

vehicle receives updated model weights, ensuring that detection capabilities evolve as attack scenarios change.

On the vehicle side, a lightweight embedded platform (e.g., Jetson Nano T206) conducts real-time inference using the cloud-trained model. It monitors incoming CAN messages and triggers alerts upon detecting anomalies. Detection outcomes and runtime metadata are uploaded to the cloud, enabling the cloud to retrain or fine-tune the model based on evolving attack strategies or environmental changes. It is important to emphasize that the vehicle is responsible solely for inference, while all training is conducted in the cloud. Model updates are performed by periodically transmitting retrained weights from the cloud to the vehicle, thereby avoiding computational bottlenecks on resource-constrained devices.

This collaborative architecture balances cloud-side computational power with vehicle-side responsiveness, ensuring scalable, low-latency, and adaptive intrusion detection. As validated in Figs. 8 and 9, this architecture ensures real-time feasibility: the cloud handles training and inference efficiently, while the vehicle-side platform achieves low-latency detection within limited memory resources. These results confirm that centralizing training in the cloud and deploying lightweight inference on vehicles is a practical and effective strategy. Based on this deployment architecture, we conduct performance evaluations of STATGRAPH and several baseline methods in both cloud and vehicle environments. Figs. 8 and 9 show the average inference time and memory usage of each method, respectively.

The results indicate that cloud devices generally offer lower inference times due to greater computational power. For

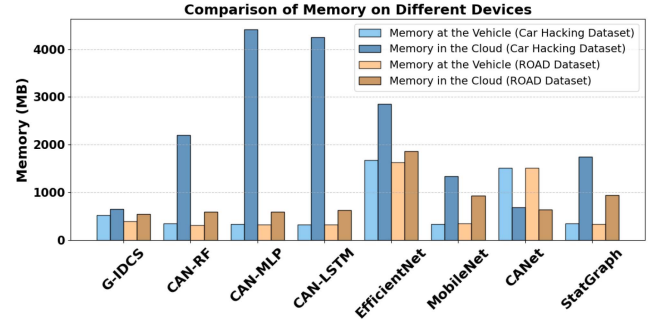


Fig. 9. Memory Comparison Evaluation.

instance, G-IDCS ML_classifier achieves low latency by summarizing 200 CAN messages into a single feature vector per window, while CAN-RF detects messages individually with minimal processing overhead. Deep learning models such as CAN-MLP and CAN-LSTM maintain stable but higher inference times due to their network complexity. Image-based models (e.g., EfficientNet, MobileNetV3, CANet) incur additional time costs from raw-to-image transformations. STATGRAPH performs competitively in the cloud but exhibits increased latency on edge devices due to time-intensive graph construction and preprocessing.

In terms of memory consumption, most models consume more resources in cloud environments. However, methods like CAN-RF, CAN-MLP, CAN-LSTM, and STATGRAPH show favorable memory footprints on embedded hardware, making them viable for real-world deployment. Notably, STATGRAPH successfully meets real-time inference requirements under resource constraints, demonstrating its practical applicability.

To further reduce communication and computation overhead, the current implementation deploys the IDS on bypass-monitoring devices connected to the CAN bus rather than directly on ECUs. This design balances detection efficiency with system safety and scalability.

In summary, our vehicle-cloud collaborative deployment aligns with real-world in-vehicle environments, allowing STATGRAPH to deliver fine-grained, real-time intrusion detection while leveraging cloud-side computational advantages for model training and continuous adaptation. This division of labor enables high detection accuracy and low latency, while ensuring scalability and deployability under constrained in-vehicle resources.

G. Detection Performance

The effectiveness of STATGRAPH is evaluated against state-of-the-art IVN intrusion detection methods. Experimental results on in-vehicle equipment demonstrate that STATGRAPH outperforms competing approaches in detecting anomalies across both datasets.

Tables VI and VII report the Accuracy, Precision, Recall, F1-score, and *Window Size/Label Number* for all evaluated methods on the Car Hacking dataset and ROAD dataset. The *Window Size/Label Number* metric reflects detection granularity

TABLE VI
TEST RESULTS OF EACH METHOD FOR DIFFERENT ATTACKS IN CAR HACKING DATASET

Dataset	Attacks	Methods	Accuracy	Precision	Recall	F1-Score	Window Size/ Label Number
Car Hacking Dataset	DoS Attack	Graph-based IDS [11]	0.432	0.766	0.387	0.514	1200/1
		G-IDCS TH_classifier [12]	0.3096	1.0000	0.3096	0.4729	200/1
		STATGRAPH	0.9980	0.9953	0.9721	0.9836	50/50
	Fuzzy Attack	Graph-based IDS [11]	0.458	0.761	0.426	0.546	1200/1
		G-IDCS TH_classifier [12]	0.3495	1.0000	0.3495	0.5180	200/1
		STATGRAPH	0.9970	0.7804	0.8392	0.8088	50/50
	Drive GEAR Attack	Graph-based IDS [11]	0.608	0.721	0.594	0.651	1200/1
		G-IDCS TH_classifier [12]	0.4379	1.0000	0.4379	0.6090	200/1
		STATGRAPH	0.9980	0.9996	0.8920	0.9427	50/50
	RPM Attack	Graph-based IDS [11]	0.899	0.971	0.856	0.910	1200/1
		G-IDCS TH_classifier [12]	0.4592	1.0000	0.4592	0.6294	200/1
		STATGRAPH	0.9980	0.9932	0.9355	0.9635	50/50
	Mixing Fabrication Attack	EfficientNet [26]	0.0688	0.2526	0.8015	0.3841	54/1
		MobileNet [27]	0.2815	0.2474	0.8485	0.3831	3/1
		CANet [16]	0.9050	0.6447	0.9725	0.7754	54/1
		G-IDCS ML_classifier [12]	0.9269	0.8759	0.8711	0.8735	200/1
		CAN-RF [28]	0.9733	0.9807	0.8208	0.8303	60/60
		CAN-MLP [28]	0.9314	0.9222	0.6731	0.7782	100/100
		CAN-LSTM [29]	0.9415	0.7521	0.8672	0.8055	100/100
		STATGRAPH	0.9940	0.9275	0.9533	0.9403	50/50

TABLE VII
BEST TEST RESULTS OF EACH METHOD FOR DIFFERENT ATTACKS IN ROAD DATASET

Dataset	Attacks	Methods	Accuracy	Precision	Recall	F1-Score	Window Size/ Label Number
ROAD Dataset	Correlated signal Masquerade Attack	Graph-based IDS [11]	0.540	0.450	0.818	0.581	1200/1
		G-IDCS TH_classifier [12]	0.6922	1.0000	0.6922	0.8181	200/1
		STATGRAPH	0.9990	0.9944	0.9976	0.9960	400/400
	Max Speedometer Masquerade Attack	Graph-based IDS [11]	0.625	0.143	0.556	0.227	1200/1
		G-IDCS TH_classifier [12]	0.3806	1.0000	0.3806	0.5513	200/1
		STATGRAPH	0.9618	0.9627	0.9618	0.9622	400/400
	Reverse Light Off Masquerade Attack	Graph-based IDS [11]	0.603	0.448	0.591	0.510	1200/1
		G-IDCS TH_classifier [12]	0.4455	1.0000	0.4455	0.6164	200/1
		STATGRAPH	0.9551	0.9122	0.9551	0.9332	400/400
	Reverse Light On Masquerade Attack	Graph-based IDS [11]	0.528	0.215	0.469	0.295	1200/1
		G-IDCS TH_classifier [12]	0.4512	1.0000	0.4512	0.6218	200/1
		STATGRAPH	0.9553	0.9132	0.9553	0.9338	400/400
	Max Engine Coolant Temp Masquerade Attack	Graph-based IDS [11]	0.830	0.111	1.000	0.200	1200/1
		G-IDCS TH_classifier [12]	0.1333	1.0000	0.1333	0.2353	200/1
		STATGRAPH	0.9955	0.9910	0.9955	0.9933	400/400
	Mixing Masquerade Attack	EfficientNet [26]	0.9699	0.5281	0.5232	0.5236	3/1
		MobileNet [27]	0.9637	0.5228	0.5084	0.5132	3/1
		CANet [16]	0.9459	0.4111	0.3292	0.3585	3/1
		G-IDCS ML_classifier [12]	0.7281	0.3115	0.2672	0.2811	200/1
		CAN-RF [28]	0.9699	0.4141	0.3429	0.3752	60/60
		CAN-MLP [28]	0.8341	0.6970	0.4181	0.5227	100/100
		CAN-LSTM [29]	0.6741	0.9577	0.6354	0.7639	100/100
		STATGRAPH	0.9897	0.9818	0.9897	0.9858	400/400

by measuring the ratio between the number of messages within a detection window and the number of classification decisions produced. Traditional coarse-grained methods assign one label per window regardless of its size, leading to high ratios (e.g., 100/1). In contrast, STATGRAPH performs detection at the message level, achieving a 1/1 ratio that represents significantly finer granularity.

As the nature of binary detection methods, Graph-based IDS [11] and G-IDCS TH_classifier [12] are fail to set consistent thresholds for multiple attack categories, and thus we only place the results of those multi-classification methods when detecting the Mixing Fabrication Attack and Mixing Masquerade Attack.

On the relatively simpler Car Hacking dataset, STATGRAPH achieves near-perfect results, with 99.40% accuracy and 94.03% F1-score. In contrast, the ROAD dataset poses a greater

challenge due to its stealthy masquerade attacks that preserve the statistical properties of normal traffic while subtly altering payload values. Despite this, STATGRAPH maintains strong performance, reaching 97.91% accuracy and 97.46% F1-score.

This robustness is attributed to the complementary roles of the TCG and CRG. The TCG captures long-term statistical behavior by aggregating messages with the same CAN ID, enabling detection of anomalies in transmission periodicity. The CRG complements this by modeling short-term contextual structure: it preserves the original temporal sequence of messages and connects those with similar IDs, making it sensitive to abrupt changes in payload distributions and inconsistencies among correlated signals within data field. These two perspectives jointly enhance the model's ability to identify both global and localized disruptions.

Crucially, the integration of TCG-based timing anomalies and CRG-reflected payload inconsistencies provides a rich and interpretable basis for anomaly detection. Variations in message periodicity, sudden signal deviations, and unexpected changes in signal correlation patterns all serve as interpretable cues for distinguishing normal from malicious behavior. Together, these mechanisms support fine-grained, message-level detection, contributing to both the high accuracy and the practical explainability of STATGRAPH.

Furthermore, baseline model generalizability varies considerably. For instance, EfficientNet and MobileNetV3 perform well on the ROAD dataset but poorly on the Car Hacking dataset, whereas CAN-RF, CAN-MLP, CAN-LSTM, and CANet exhibit the opposite trend. In contrast, STATGRAPH consistently achieves the highest Accuracy and F1-score on both datasets, demonstrating superior adaptability and robustness. The *Window Size/Label Number* metric further highlights a key advantage of STATGRAPH over existing graph-based approaches. While most methods operate at the window or segment level, STATGRAPH generates predictions for each message individually. This fine-grained detection enables earlier threat identification and more precise localization, which is especially valuable in time-sensitive and safety-critical IVN environments.

H. Fine-Grained Potential Exploration

Previous studies typically labeled an entire segment of consecutive CAN messages as an attack sample if it contained at least one injected message, while segments without injections were labeled as normal [10]. However, aside from a few graph-based methods [11], [12], prior work has not provided a clear rationale for the choice of sample size.

To address this gap, we define sample size criteria based on CAN bus transmission rates and latency requirements, referencing application scenarios in *Intelligent Connected Vehicles* and *Industrial Control Internet*. Industrial automation and remote driving generally require latency below 10ms. Given a CAN bus speed of 125kbps to 1Mbps and a maximum extended frame length of 150 bits, the bus can transmit at least 8 messages and on average 34 messages every 10ms at 500kbps. Based on this, we define three recognition granularity intervals: $0 \sim 8$, $8 \sim 34$, and $34 \sim +\infty$, to evaluate the performance of coarse-grained methods across varying levels of granularity. Corresponding detection window sizes are set to 3, 27, and 54 for comparison. STATGRAPH is evaluated against prior methods using F1-score and IG under these settings, as reported in Table VIII. Notably, STATGRAPH performs detection at the per-message level regardless of the window size $N = (3, 27, 54)$, yielding stable and consistent results across all intervals.

It is evident that STATGRAPH achieves the highest F1-score and IG values on both datasets. On the Car Hacking dataset, among the remaining models, CANet delivers the best performance when $N = 54$, suggesting that it relies on a larger detection window to accumulate sufficient contextual information for effective classification. On the ROAD dataset, where normal data accounts for over 90% of the samples, many models tend to classify most messages as normal. Compared to other

TABLE VIII
TEST RESULTS OF EACH METHOD IN MIX ATTACKS (REAL-TIME DETECTION:
 $N = 3$, SEMI-REAL-TIME DETECTION: $N = 27$ AND OFFLINE DETECTION:
 $N = 54$)

Method	Window Size	F1-Score (Car Hacking)	IG (Car Hacking)	F1-Score (ROAD)	IG (ROAD)
EfficientNet	$N = 54$	<u>0.3841</u>	<u>0.0128</u>	0.5236	0.4664
	$N = 27$	0.1382	0.0054	0.3348	0.4957
	$N = 3$	0.1215	0.0041	<u>0.5236</u>	<u>0.9811</u>
MobileNetV3	$N = 54$	0.1018	0.0046	0.2861	0.4384
	$N = 27$	0.1109	0.0055	0.3411	0.4657
	$N = 3$	<u>0.3831</u>	<u>0.2815</u>	<u>0.5132</u>	<u>0.9285</u>
CANet	$N = 54$	<u>0.7754</u>	<u>0.8996</u>	0.1060	0.4664
	$N = 27$	0.5755	0.7910	0.1100	0.4942
	$N = 3$	0.7023	0.7229	<u>0.3585</u>	<u>0.9261</u>
STATGRAPH	$N = 54$	0.9403	0.9940	0.9858	0.9897
	$N = 27$	0.9403	0.9940	0.9858	0.9897
	$N = 3$	0.9403	0.9940	0.9858	0.9897

methods, STATGRAPH consistently achieves higher IG values and F1-scores across all window sizes on both datasets, demonstrating its superior accuracy and robustness in the presence of imbalanced data distributions.

I. Ablation Experiment

Within STATGRAPH, the TCG and CRG are designed to capture distinct structural properties of CAN traffic, grounded in different theoretical assumptions. The TCG, proposed in the Global Feature Fusion Module, aggregates all messages with the same CAN ID to form node-level statistical features such as frequency and entropy. This global representation captures long-range behavioral patterns across messages sharing the same ID that is effective for detecting distributional shifts caused by attacks. In contrast, the CRG, constructed in the Association Enhancement Module, retains each message as an individual node and connects them based on temporal adjacency and ID similarity. By preserving both temporal order and message-level granularity, the CRG captures fine-grained structural dynamics and enables the detection of localized disruptions that may be overlooked by aggregated models. This dual-graph architecture allows STATGRAPH to jointly encode global distributional context and localized temporal structure, offering a comprehensive representation of communication dependencies in IVNs.

To assess the individual contributions of global feature fusion and temporal adjacency enhancement, we conduct an ablation study. In the Node ablation setting (excluding the TCG), only 8-dimensional payload vectors are used as node features, excluding the fused global attributes. This allows us to evaluate the impact of global feature fusion on node representation. In the Edge ablation setting (excluding the CRG), the adjacency matrix generated by the temporal association module is replaced with a unit matrix, thereby removing temporal dependencies and enabling assessment of correlation modeling.

The effectiveness of this design is validated by the ablation results shown in Fig. 10. Although the Edge ablation model achieves slightly higher accuracy and F1-score on the Car Hacking dataset, it suffers a substantial drop in recall, indicating that temporal adjacency is critical for detecting low-frequency or stealthy attacks. The Node ablation model shows consistent performance degradation across all major metrics, underscoring

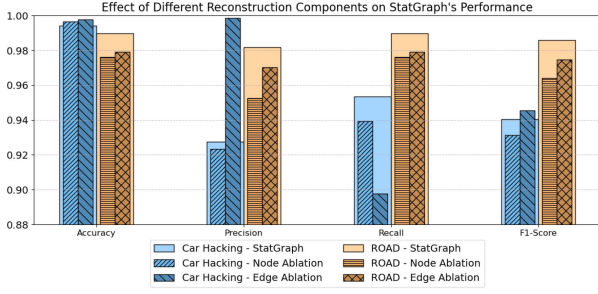


Fig. 10. Performance of different ablation components of STATGRAPH on two datasets.

the role of global feature fusion in improving node representation and detection accuracy.

Taken together, these findings confirm that STATGRAPH achieves optimal performance when both temporal correlations and global contextual features are preserved. In particular, improvements in recall and F1-score underscore the model's ability to deliver fine-grained and robust intrusion detection in complex in-vehicle network environments.

V. DISCUSSION

The experimental results demonstrate that STATGRAPH consistently outperforms existing methods on both the Car Hacking dataset and the ROAD dataset when evaluated independently. Although some alternatives achieve satisfactory performance, such as EfficientNet and MobileNetV3 on the ROAD dataset, and CANet, G-IDCS ML_classifier, CAN-RF, CAN-MLP, and CAN-LSTM on the Car Hacking dataset, STATGRAPH achieves the most robust and consistent results across both datasets.

While our TCG design shares conceptual similarities with [11], STATGRAPH introduces key advancements by incorporating the CRG and enabling fine-grained, message-level classification. In contrast to [11], which relies on window-level statistical features, STATGRAPH performs per-message detection by fusing global statistical context with local payload and signals dynamics. This allows it to capture subtle anomalies (e.g., mixing masquerade attacks) that are often missed by traditional threshold-based approaches.

In addition, STATGRAPH is deployed within a practical vehicle-cloud collaborative framework, delivering strong detection performance even in the presence of stealthy and low-frequency masquerade attacks. These architectural innovations and deployment strategies represent a substantial and novel contribution to graph-based intrusion detection for IVNs.

In real in-vehicle environments, message transmission frequencies vary with operating conditions, which may limit the practicality of fixed-window strategies. This limitation is not specific to our work but represents a broader challenge for IDS deployment. As future work, we plan to incorporate adaptive window mechanisms that adjust N dynamically based on real-time message rates. We also aim to investigate multi-scale window modeling, where detection results across different temporal resolutions are integrated to improve robustness against both bursty and stealthy attacks.

Moreover, the limitations of supervised models, particularly the reliance on labeled data, remain a key challenge. This motivates the development of self-supervised and unsupervised frameworks that can adapt to evolving threats without annotated attack data. The graph-based representation in our framework, comprising both the TCG and CRG, provides a structured input space well-suited for self-supervised objectives, including contrastive learning, masked signal prediction, and graph structure reconstruction. For instance, intrusion detection can be performed by applying perturbations to subgraphs and using graph contrastive learning to capture intrinsic features of normal behavior, allowing the model to differentiate between benign and malicious graphs without ground-truth labels. Extending STATGRAPH with such self-supervised strategies thus offers a promising direction to enhance robustness in low-label and zero-day attack scenarios.

Another important direction is addressing data imbalance, which can impair detection under sparse or skewed attack distributions. Future work may incorporate synthetic data generation or adversarial training strategies to enhance model robustness.

In addition, improving model interpretability is essential for practical deployment. Post-hoc interpretation techniques, such as GNNExplainer [35] and Integrated Gradients [36], can be employed to identify the node features (derived from the TCG) or edge structures (captured by the CRG) that contribute most significantly to anomaly classification. These insights can help practitioners assess whether the model's decisions align with domain knowledge, thereby increasing trust and enhancing the diagnostic value of intrusion detection systems in safety-critical onboard environments.

VI. RELATED WORK

Deep learning-based intrusion detection methods hold significant promise for securing IVNs, but they face critical limitations in detection granularity and effectiveness against sophisticated attacks, which constrain their practical deployment [4], [6], [9], [21], [22]. First, most existing approaches perform coarse-grained detection, identifying anomalies only at the detection window level rather than at the level of individual CAN messages [14], [15]. This lack of fine-grained resolution undermines their diagnostic reliability and limits timely threat localization. Second, as attack strategies evolve, increasingly advanced masquerade attacks have emerged. These attacks manipulate the payload content while mimicking the timing and frequency of legitimate messages, allowing them to evade detection by existing models that rely primarily on statistical or temporal patterns [5], [7], [19].

Besides, graph-based intrusion detection methods have gained traction in IVN security due to their strong abstraction capabilities in modeling interactions among heterogeneous ECUs, effectively capturing the collaborative processes and data distribution patterns of the CAN bus. However, most existing graph-based approaches focus predominantly on detecting relatively simple types of attacks [11], while the detection of advanced and stealthy masquerade attacks remains significantly underexplored. Methods such as [11], [12], [37], and [38] rely

solely on CAN IDs and overlook the critical need to learn payload variation patterns. As a result, they are ineffective against masquerade attacks that precisely replicate the statistical properties of legitimate messages but subtly manipulate payload thresholds. Furthermore, many graph-based methods adopt coarse-grained detection strategies, identifying anomalies only after large message intervals (e.g., every 1200 messages in [11] and every 200 in [12]), which fails to satisfy the real-time responsiveness required in IVN applications. In contrast to most existing single-view models [11], [12], [37], [38], STATGRAPH introduces a multi-view graph representation that captures diverse relationships among CAN messages. By integrating different graph perspectives, STATGRAPH is better positioned to detect subtle and complex intrusion behaviors with higher fidelity.

VII. CONCLUSION

Intrusion detection based on deep neural networks and graph learning has advanced IVN security, yet improving detection granularity and identifying complex attacks remains challenging. To address these limitations, we propose STATGRAPH, a fine-grained IVN intrusion detection framework that effectively detects both simple attacks (e.g., fabrication) and sophisticated ones (e.g., masquerade) by leveraging multi-view statistical graph learning on CAN messages. STATGRAPH adopts a dual-graph design in which the TCG captures global temporal and distributional patterns, while the CRG retains local adjacency and semantic consistency at the message level. Their integration through graph convolution enables cross-graph feature propagation, allowing STATGRAPH to effectively capture variations in message periodicity, payload content, and signal correlations. We further introduce the Identification Granularity (IG) metric to quantify detection resolution, providing a foundation for future research on fine-grained intrusion detection. Extensive experiments demonstrate that STATGRAPH outperforms traditional machine learning models, deep neural networks, and existing graph-based methods in terms of runtime efficiency, detection accuracy, and granularity across both the Car Hacking and ROAD datasets. Future work will explore attack traceability using fine-grained detection and develop unsupervised intrusion detection mechanisms to address class imbalance problems in real-world scenarios.

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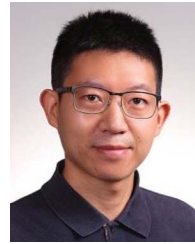
Kai Wang (Member, IEEE) received the PhD degree in communication and information systems from Beijing Jiaotong University, China, in 2014. He is a full professor with the School of Computer Science and Technology, Weihai and with the Faculty of Computing, Harbin, Harbin Institute of Technology, China. His research interests include trustworthy AI and network intrusion detection. He has published more than 40 papers in prestigious international journals, including the *IEEE Transactions on Services Computing*, *IEEE Transactions on Intelligent Transportation Systems*, *ACM Transactions on Internet Technology*, *ACM Transactions on Intelligent Systems and Technology*, etc. From 2017 to 2019, he was a postdoc researcher in computer science and technology with Tsinghua University, China. He is a member of the ACM, and a senior member of the China Computer Federation (CCF).



Qiguang Jiang received the master's degree in computer science and technology from the Harbin Institute of Technology (HIT), China. Her research interests include intelligent and efficient in-vehicle intrusion detection models.



Bailing Wang (Member, IEEE) received the PhD degree in computer architecture from the Harbin Institute of Technology (HIT), China, in 2006. He is a full professor with the Harbin Institute of Technology (Weihai) Qingdao Research Institute, Qingdao, China. His research interests include information content security and industrial control network security. He has published more than 80 papers in prestigious international journals and been selected for the China national talent plan.



Yulei Wu (Senior Member, IEEE) received the BSc (1st class hon.) degree in computer science and the PhD degree in computing and mathematics from the University of Bradford, United Kingdom. He is an associate professor with the Faculty of Engineering and the Bristol Digital Futures Institute, University of Bristol, U.K. His research mainly focuses on network digital twins, native AI networks and systems, edge AI, and trustworthy AI. He has published more than 10 authored/edited monograph books, and more than 150 peer-reviewed research papers in prestigious international journals and conferences.

He serves as an associate editor of the *IEEE Transactions on Network and Service Management* and *IEEE Transactions on Network Science and Engineering*, as well as an editorial board member of the *Computer Networks*, *Future Generation Computer Systems*, and *Nature Scientific Reports at Nature Portfolio*. He is chairing an IEEE Special Interest Group (SIG) on Ethical AI for future networks and digital infrastructure. He is a senior member of the ACM.



Hongke Zhang (Fellow, IEEE) received the PhD degree in communication and information system from the University of Electronic Science and Technology of China, Chengdu, China, in 1992. He is currently a professor with the School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing, China, where he currently directs the National Engineering Center of China on Mobile Specialized Network. His current research interests include architecture and protocol design for the future Internet and specialized networks. He currently serves as an

associate editor of the *IEEE Transactions on Network and Service Management* and *IEEE Internet of Things Journal*. He is an academican of China Engineering Academy.